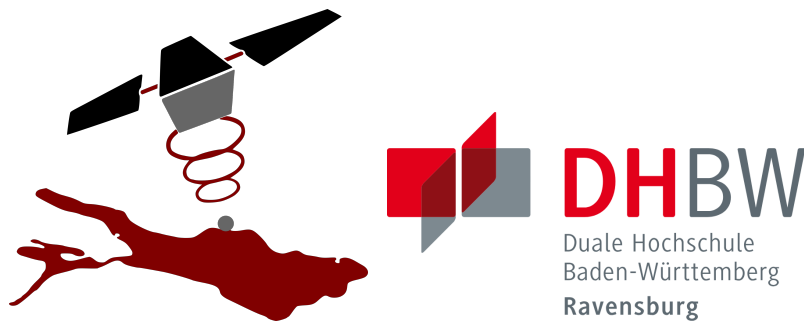


Development Process

for

Development of a telemetry data link for the
ERWIN cubesat



Project:	HW/SW Projekt TSL - 0004
Date:	October 23, 2025
Version:	1.0
Description:	Documentation of the Development Process for this project

Revision History

Name	Date	Description	Version
Matthias Fuchs	17.10.2025	First Version	1.0
Matthias Fuchs	23.10.2025	Add WPD, EPIC, STORY	2.0
Matthias Fuchs	23.10.2025	Add Naming and Versioning	3.0

Contents

1	Introduction	1
1.1	Purpose	1
1.2	Scope	1
2	Terms and Definitions	2
2.1	V-Model	2
2.2	V-Model-XT	2
2.3	Iterative approach	2
3	Process Definition	3
3.1	Structure	3
3.1.1	User Layer	3
3.1.2	High Level Layer	3
3.1.3	Low Level Layer	4
3.1.4	Coding Layer	4
3.1.5	Work Packet Description	4
3.1.6	Epic	5

	3.1.7	Story	5
3.2	Roles		5
	3.2.1	Development Team	5
	3.2.2	Scrum Master	6
	3.2.3	Feel Well Manager	6
	3.2.4	Implementer	6
	3.2.5	Tester	6
	3.2.6	Git Administrator	7
	3.2.7	Standard Expert	7
	3.2.8	Note Taker	7
	3.2.9	Comunication Manager	7
4	Process Layout		9
5	Other Definitions		9
	5.1	Standards and Guidelines Naming and Versioning	9

1 Introduction

1.1 Purpose

This document will define the development process that the project team has decided upon. This includes the definition of the tailored V-Model used in this project, as well as the role definitions and customisation.

1.2 Scope

This Standard applies to all development done for the development of the data link layer of the TM link of ERWIN.

2 Terms and Definitions

2.1 V-Model

The classical V-model is structured in the shape of a V. The development process closely follows this shape. On the left-hand side of the V, the downward path, the software design and requirements become increasingly precise until the bottom of the V is reached. Here, the design is implemented in code. Testing is situated on the right-hand side of the V. Testing is always conducted against the corresponding layer on the left-hand side.

2.2 V-Model-XT

The V-Model-XT aims to improve the classical V-Model and tailor it for German military projects, even though it is now used for many more projects. The classical V-Model is expanded to include an iterative approach, enabling a prime contractor to include subcontractors in the process. These subcontractors also follow the V-Model-XT and may include subcontractors of their own.

The V-Model-XT also defines a large number of roles and positions, some of which will be used in this development process.

2.3 Iterative approach

The Iterative approach in general defines many different Development Processes that can be highly tailored to the needs of a project. Generally a project will be improved in iterative stages. With this Rapid Prototyping approach different projects parts may progress at their own pace.

3 Process Definition

3.1 Structure

STRU-REQ-1 The basic structure shall follow the Classical V-Model

STRU-REQ-2 The V-Model shall be tailored to fit the needs of the Project

STRU-REQ-3 The V-Model shall be subdivided into 4 Horizontal Layers

STRU-REQ-4 The V-Model shall be mapped to an iterative (scrum like) approach in the Lower 2 Layers

3.1.1 User Layer

USER-REQ-1 This Layer shall define the requirements provided by the end Users

USER-REQ-2 This Layer shall define the final accepted tests required by the end User

USER-REQ-3 In this Project the end User shall be the SeeSat e.V. for the Project Erwin

3.1.2 High Level Layer

HILE-REQ-1 This Layer shall define High Level/System requirements

HILE-REQ-2 This Layer shall define System Integration tests

HILE-REQ-3 A High Level Requirement shall correspond to a Epic, as defined in iterative development

HILE-REQ-4 The High Level Requirements shall be linked to the User Layer (tracing)

3.1.3 Low Level Layer

- LOLE-REQ-1** This Layer shall define the Low Level Requierments
- LOLE-REQ-2** This Layer shall define Unit tests
- LOLE-REQ-3** Each Unit tests shall be used to test one Low Level Requierment
- LOLE-REQ-4** The Low Level Requierments shall define a detailed design (acording to DO-178C)
- LOLE-REQ-5** The Low Level Requierments shall be directly mapped to code
- LOLE-REQ-6** The Low Level Requierments shall be linked to High Level Requierments
- LOLE-REQ-7** The Low Level Requierments shall be changed if and only if there is logical Mistake found in the Implementation or the Unit Testing of the coresponding Low Level Requierment
- LOLE-REQ-8** The Low Level Requierments shall correspond to a Story, as defined in iterative developement

3.1.4 Coding Layer

- CODE-REQ-1** This Layer shall define only the Final implementation of the Low Level Requierments
- CODE-REQ-2** The Coding Layer shall corespond to a Ticket, as defined in iterative developement
- CODE-REQ-3** It Shall be forbidden that a person that implements a low level requierment in the Coding layer shall also tests their own implementation (more in the roles part)

3.1.5 Work Packet Description

- WDP-REQ-1** This data is provided by the SeeSat e.V.

WDP-REQ-2 The Work Packet Description (WPD) should define exactly how the SeeSat intends to use the System(s) to be developed.

WDP-REQ-3 The WDP should define Dokumentation requierments

WDP-REQ-4 The WDP should define deliverables requiered by the SeeSat e.V.

3.1.6 Epic

EPIC-REQ-1 A Epic shall define a large Subsection/Subsystem of the WPD.

EPIC-REQ-2 A Epic should have a work load of about 8 Sprints.

EPIC-REQ-3 A Epic should be used to represend a whole subsystem.

3.1.7 Story

STOR-REQ-1 A Story shall define a small piece of work.

STOR-REQ-2 A Story should have a work load of about 1 Sprint.

STOR-REQ-3 A Story should directly map to a small subset of Low Level Requierments

3.2 Roles

The Clasical V-Model and V-Model-XT define many roles. Since this project defines some custom roles and redefines some others these Role definitions shall be done in this Chapter

3.2.1 Developement Team

DEVE-REQ-1 Every one on the Team shall be part of the Development Team

DEVE-REQ-2 Every one on the Development Team shall have one Vote if it comes to a internal discussion

DEVE-REQ-3 Every one on the Development Team shall fulfill one or more of the roles defined in this chapter

3.2.2 Scrum Master

SCRU-REQ-1 Not a single person shall be the Scrum Master but everyone should be it for a few weeks

SCRU-REQ-2 The Scrum Master shall organize the needed meetings

3.2.3 Feel Well Manager

FEEL-REQ-1 The Feel Well Manager shall be responsible for bringing something nice to longer meetings

FEEL-REQ-2 Longer meetings shall be defined by the Team, e.g. if it is known in advance that a meeting will take longer

FEEL-REQ-3 At the end of a Meeting it shall be decided who the Feel Well Manager shall be for the next meeting

3.2.4 Implementer

IMPL-REQ-1 This Person shall be responsible for implementing the Low Level Requirements in Code

IMPL-REQ-2 There shall not be a Singular Person on the Team that does the implementation

IMPL-REQ-3 The Implementer shall not Unit-Test their own code (Self testing of Code shall be permitted but only for self validation not for requirement Testing)

3.2.5 Tester

TEST-REQ-1 This Person shall be responsible for writing Test Cases for Low Level Requirements

3.2.6 Git Administrator

GIT-REQ-1 One Person one the team shall be the Git Admin

GIT-REQ-2 The Git Admin shall be responsible for setting up the Gitlab workspace for everyone

GIT-REQ-3 The Git Admin shall watch the CI/CD pipelines

GIT-REQ-4 If a CI/CD pipeline fails the Git Admin shall adminster a ticket to the Person responsible for that pipeline

3.2.7 Standard Expert

STAN-REQ-1 For every CCSDS Standard (or another Standard if needed) there shall be one or two Standard Expert

STAN-REQ-2 A Standard Expert shall be responsible for reading their standard precisely

STAN-REQ-3 A Standard Expert shall be able to help other team members with the Standard if they have detailed questions

STAN-REQ-4 The Standard Expert shall never be the singular source of information if in doubt the Standard its self shall allways be consilutated

3.2.8 Note Taker

NOTE-REQ-1 For every meeting there shall be Person deticated to taking meeting notes

NOTE-REQ-2 This Note Taker shall not always be the same person, it has been defined to rotate this role

3.2.9 Comunication Manager

COM-REQ-1 The Comunication Manager shall be responsible for all external E-Mail Comunication

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- COM-REQ-2** The Communication Manager shall formulate an update E-Mail every week
- COM-REQ-3** The weekly update E-Mail shall be send out for internal review not later than Saturday
- COM-REQ-4** The weekly update E-Mail shall be send out to the Customer at Sundays
- COM-REQ-5** The Contents of a weekly update E-Mail shall be discussed in the weekly meetings

4 Process Layout

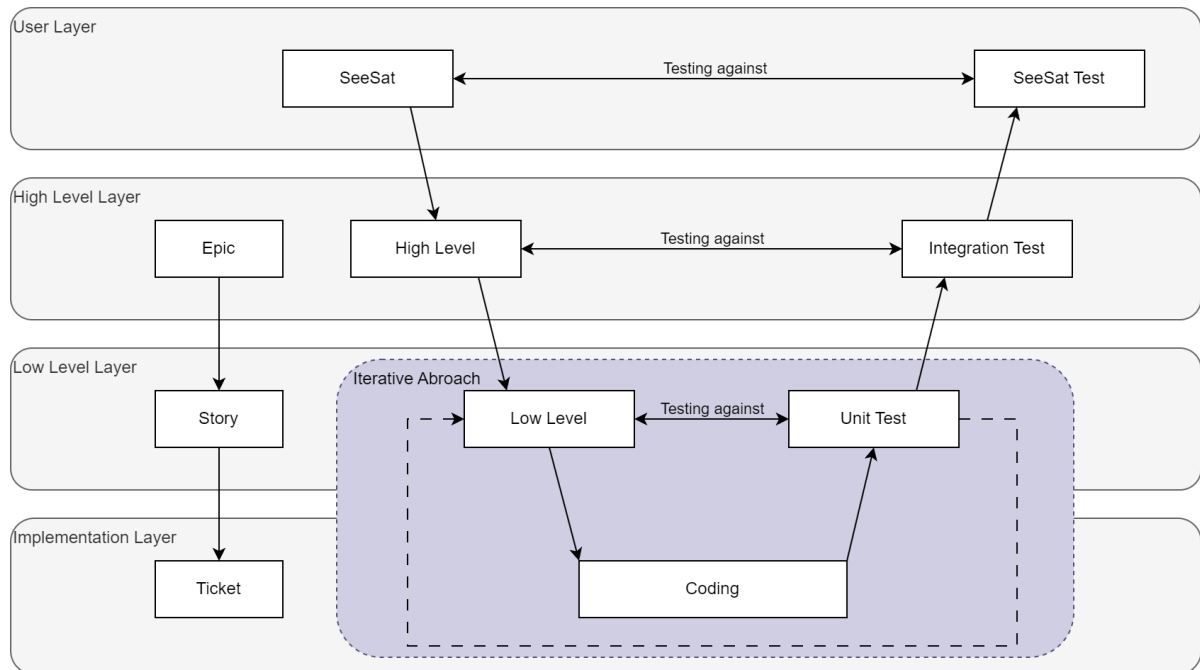


Figure 4.1: Tailored Development process after the V-Model

5 Other Definitions

5.1 Standards and Guidelines Naming and Versioning

SaG-REQ-1 All Standards and Guidelines written in for this Project shall be saved to SeeSat Nextcloud

SaG-REQ-2 The Local Path to these Standards and Guidelines shall be $\{\text{ProjectRoot}\}/\text{Documents}/\text{Standardization}/\{\text{Standard Name}\}/$

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- SaG-REQ-3** The Current version of a Standard or Guideline shall be saved in the directory as defined in SaG-REQ-2
- SaG-REQ-4** All old versions shall be saved in
\${Standard Name}/Version_History/
- SaG-REQ-5** Every Standard and Guideline version saved in the Version History shall use this naming scheme
\${Standard Name}_\${VersionNumber}.pdf
- SaG-REQ-6** The Current version of the Standards and Guidelines shall use this naming scheme \${Standard Name}.pdf
- SaG-REQ-7** The first release of a Standard or Guideline shall be the V1.0
- SaG-REQ-8** The next major change of a Standard or Guideline shall be the VX.0, where X is one bigger than the previous Version.
- SaG-REQ-9** a Major change may be the addition of new chapter(s) or the reworking of large parts of the document.
- SaG-REQ-10** The next minor change of a document shall be the VX.Y where Y is one bigger than the previous minor version, X shall not change between minor changes.
- SaG-REQ-11** a Minor change may be a change of a small number of requirements or definitions.
- SaG-REQ-12** Gramatical and Typographical changes may be applied to the same version without change of Version number, this shall only be done if and only if these changes do not change the meaning of the text.